ENTERPRISE AI PLATFORM • 2026 CAPSTONE EDITION

SENTILINK ENTERPRISE

Autonomous Book Review Sentiment Analysis & Reader Opinion Mining

A production 3-Tier MVC architecture unifying multi-method NLP text mining, real-time opinion extraction, and rigorous IT project delivery governance.

ELECTIVE VI A

Enterprise Systems

SOA REST, Master Data, RBAC & 5-Yr TCO

ELECTIVE IV B

Advanced Text Analytics

VADER, RAKE, K-Means & Apriori Rules

IT PROJECT MGMT

ITPM Governance

WBS, 16-Wk Plan, PERT CPM & ₹55k Cost

Bharati Vidyapeeth Deemed University (BVDU DET Pune) • Computer Science & Business Systems

EXECUTIVE OVERVIEW

One Unified Platform • Three Core Subject Pillars

Bridging academic theory with enterprise production engineering across ES, ASTMA, and ITPM.

ASTMA

Text & Media Analytics

- **Unsupervised Topics**
  K-Means discovery of 4 latent themes
- **Valence Scoring**
  Dual VADER + Supervised Naive Bayes
- **Keyphrase Mining**
  RAKE word degree to frequency ratio
- **Network Analytics**
  Bipartite reader-book social graph

ES

Enterprise Systems

- **3-Tier Architecture**
  Presentation, SOA API, Relational DB
- **Master Catalog**
  36 books across 35 distinct genres
- **RBAC Security**
  Admin moderation vs Customer workflows
- **Economic Modeling**
  5-Year TCO: COTS vs Bespoke build

ITPM

Project Delivery

- **Scope & WBS**
  5 Work packages & 15 discrete tasks
- **Schedule Control**
  16-Week lifecycle & 7 stage-gates
- **CPM / PERT**
  Probabilistic expected duration & variance
- **Cost Governance**
  Bottom-up ₹55,000 budget tracking

124+ Stored Reviews
Validated Corpus

35 Master Genres
Literature Diversity

10 Domain Aspects
Opinion Dimensions

₹55,000 Budget
Strict Financial Cap

ENTERPRISE SYSTEMS

3-Tier MVC Architecture & Service-Oriented Framework

Clean separation of concerns with asynchronous REST contracts and relational ACID transactions.

PRESENTATION LAYER (CLIENT TIER)

Tailwind CSS 3.0 • Vanilla JS (ES6+) • Chart.js 4.4 • Lucide Icons

Responsive multi-view dashboard, live NLP typing feedback, category filters & modal inspectors.

BUSINESS LOGIC LAYER (APPLICATION TIER)

FastAPI SOA Controller • Python 3.13 • Pydantic V2 • nlp_engine.py

20+ REST endpoints, VADER valence calculation, RAKE keyphrase extractor, K-Means & Apriori.

DATA PERSISTENCE LAYER (DATABASE TIER)

SQLite 3.45 • B-Tree Indexing • Foreign Key Constraints • WAL Journaling

Relational schema: books, reviews, users, audit_logs. Sub-5ms query lookups.

Enterprise System Criteria

Key attributes ensuring production viability:

- **Shared Single Source of Truth**
  Catalog, reviews, and NLP predictions live on on e synchronized platform.
- **Strict ACID Transactionality**
  Atomic review submission with audit log triggers prevents state drift.
- **Role-Based Access Control**
  Customer review submission vs Admin review moder ation & book management.
- **Microservice Ready SOA**
  Headless REST design allows hot-swapping NLP bac kends without UI disruption.

ASTMA PIPELINE

6-Stage Asynchronous Text Mining & NLP Workflow

Sequential enrichment transforming raw unstructured customer reviews into actionable intelligence.

01
INGESTION
Multi-Source Feed
Amazon, Kaggle & User Submitted Text

02
CLEANING
Noise Strip
URLs, HTML tags, speci al symbols & case fold

03
TOKENIZE
Linguistic Split
Atomic word tokens & s topword elimination

04
VALENCE
VADER Scoring
Compound [-1, +1] & pu nctuation boost

05
ASPECTS
Domain Regex
10 Literary aspects & RAKE keyphrase tags

06
EXECUTIVE
BI Dashboard
Real-time KPIs, CSAT g auge & visual charts

Architectural Pipeline Rationale

1. Deterministic Sequential Ingestion: Raw reviews must undergo schema validation prior to tokenisation.
2. VADER Raw Context Preservation: Unlike TF-IDF, VADER requires punctuation ('!') and capitalization for intensity scoring.
3. Aspect-Polarity Decoupling: Extracting domain features independently allows negative sentiment to isolate specific flaws.

ASTMA UNIT II

Sentiment Classification: Rule-Based VADER vs Supervised ML

Production deployment of lexical heuristics benchmarked against TF-IDF Naive Bayes classification.

PRODUCTION ENGINE

VADER Lexicon & Rule Engine

Gold-standard valence dictionary optimized for micro-text and consumer reviews.

- **Punctuation Intensity**
  Repeated exclamation marks ('Must read!!!') increase polarity weight.
- **Capitalisation Booster**
  ALL-CAPS words ('AMAZING worldbuilding') amplify positive valence.
- **Degree Adverb Modifiers**
  'deeply touching' boosts score; 'slightly predictable' dampens score.
- **Polarity Flipping Negation**
  'not gripping' or 'never connected' inverts positive word polarity.
- **Contrastive 'BUT' Conjunction**
  Shifts emotional weight to the trailing clause: 'Pacing was slow, BUT loved it.'

ACADEMIC BENCHMARK

TF-IDF + Multinomial Naive Bayes

Supervised classification model trained on n-gram frequency matrices.

Accuracy	Precision	Recall	F1-Score
89.2%	0.884	0.879	0.881

Core Trade-off Analysis:

- **Explainability**
  VADER is 100% white-box; every word delta can be traced in real time.
- **Latency Footprint**
  VADER executes in <2ms per review; zero model loading or GPU dependency.
- **Domain Sensitivity**
  Supervised ML excels on genre jargon but suffers from out-of-vocabulary drift.

ASTMA UNIT III

Plot & Story

Narrative arc, suspense, mystery, storyline

Writing Style

Prose quality, poetic voice, author craft

Character Dev.

Protagonist growth, flawed motives, depth

Pacing

Page-turner tempo, slow middle, dragged chapters

Emotional Depth

Heartwarming, tear-jerker, vulnerability

Worldbuilding

Sci-fi lore, fantasy realm, immersion

Dialogue

Witty banter, realistic conversation, voice

Ending & Climax

Shocking twist, cliffhanger, resolution

Originality

Unique premise, trope subversion, fresh ideas

Price & Value

Paperback quality, audio length, affordability

LIVE SYSTEM INFERENCE PREVIEW

“The writing style is breathtaking and the emotional depth kept me hooked through chapter ten, however the pacing dragged in the middle and the character development felt rushed.”

DETECTED ASPECTS: Writing Style (+0.82) · Emotional Depth (+0.75) · Pacing (-0.45) · Character Dev. (-0.38)

RAKE KEYPHRASES: Breathtaking Writing Style (9.0) · Rushed Character Development (9.0) · Emotional Depth (4.0)

ASTMA UNIT IV

Unsupervised Machine Learning & Latent Theme Mining

K-Means TF-IDF vector clustering ($K=4$) and market-basket Apriori association rule discovery.

UNSUPERVISED CLUSTERING

K-Means Latent Theme Discovery

Discovers natural semantic groupings without supervision (K=4 in TF-IDF space).

- **Cluster 1: Cosmic Sci-Fi & Worldbuilding**
  77.4% of corpus • 'science, fiction, atmospheric, universe'
- **Cluster 2: Emotional Drama & Resonance**
  9.7% of corpus • 'pacing, tone, vulnerability, touching'
- **Cluster 3: Behavioral Science & Habit**
  8.9% of corpus • 'prose, habits, mindset, behavioral, daily'
- **Cluster 4: Psychological Suspense & Thriller**
  4.0% of corpus • 'mystery, twist, psychological, climax'

ASSOCIATION MINING

Apriori Co-occurrence Rules

Identifies high-confidence aspect relationships: Antecedent → Consequent.

- **Emotional Depth → Writing Style**
  Support: 31.5% • Confidence: 82.4% • Lift: 1.45
- **Pacing Friction → Ending Disappointment**
  Support: 18.2% • Confidence: 74.1% • Lift: 1.88
- **Worldbuilding → Originality Acclaim**
  Support: 24.6% • Confidence: 89.0% • Lift: 1.62
- **Price Critique → Negative CSAT**
  Support: 14.3% • Confidence: 91.2% • Lift: 2.14

ENTERPRISE SYSTEMS

Role-Based Access Control & Master Data Management

Operational enterprise integrity separating administrative moderation from customer interactions.

SECURITY ARCHITECTURE

RBAC Security & Audit Trails

Enforces strict authorization boundaries with transactional audit logging.

ADMIN ROLE **Product registration, delete reviews, ITPM plan view**

- Access to PERT crashing tools, Scrum boards, and ERP TCO projections.
- Review moderation authority and direct database deletion rights.

CUSTOMER ROLE **Catalog browsing, review submission, live NLP feedback**

- Live NLP feedback typing card (instant VADER score & RAKE tags).
- Read-only access to community reviews and individual book metrics.

ERP MATERIALS MODULE

Master Catalog & Data Models

Enterprise Materials Master: 36 verified books spanning 35 genres.

- **Entity Integrity**
  Foreign key constraints link reviews to verified book IDs and user IDs.
- **B-Tree Optimization**
  Indexed queries on genre, rating, and sentiment execute in <5ms.
- **Audit Trail Ledger**
  Every login, submission, and moderation writes to audit_logs table.
- **Batch ETL Capabilities**
  Paginated streaming handles 100k+ reviews without memory exhaustion.

ENTERPRISE SYSTEMS

5-Year Total Cost of Ownership (TCO): COTS vs Bespoke

PI NLP

COMMERCIAL OFF-THE-SHELF (COTS)

5-Year Cumulative: ₹1,85,000

- High recurring SaaS license fees per seat per year.
- Rigid customization lock-in; cannot tune internal NLP algorithms.
- Expensive third-party connector fees for external CRM/ERP sync.

SENTILINK BESPOKE BUILD (CHOSEN)

5-Year Cumulative: ₹82,000 (56% Savings)

- Upfront CapEx of ₹55,000 with nominal annual cloud hosting (₹6,000/yr).
- Complete ownership of NLP intellectual property and fine-tuning rules.
- Breakeven achieved by Month 14 of commercial deployment.

Cumulative Cost Projection Table (₹ INR)

Cost Category	Year 1 (Capex)	Year 2	Year 3	Year 4	Year 5	5-Yr Total
COTS SaaS Package (e.g. Brand24 / Sprinklr)	₹45,000	₹35,000	₹35,000	₹35,000	₹35,000	₹1,85,000
Sentilink Bespoke Architecture (Built)	₹55,000	₹6,000	₹7,000	₹7,000	₹7,000	₹82,000
Net Enterprise Cost Benefit (Savings)	- ₹10,000	+ ₹29,000	+ ₹28,000	+ ₹28,000	+ ₹28,000	+ ₹1,03,000

IT PROJECT MANAGEMENT

Work Breakdown Structure (WBS) & 16-Week Schedule

Structured project delivery decomposing capstone scope into 5 work packages and 15 trackable activities.

WBS SCOPE HIERARCHY

Work Breakdown Structure

- **WP 1.0: Initiation & Architecture**
  Scope charter, literature survey, 3-tier MVC blueprint
- **WP 2.0: Data Ingestion & Storage**
  SQLite schema design, Kaggle ETL pipeline, data cleaning
- **WP 3.0: Core NLP Engineering**
  VADER rule adaptation, RAKE extractor, K-Means clustering
- **WP 4.0: Dashboard & Web UI**
  Chart.js analytics, role toggles, live typing inference card
- **WP 5.0: QA, Testing & Deployment**
  Unit test suites, syllabus report generation, UAT sign-off

TIMELINE & STAGE-GATES

16-Week Delivery Lifecycle

- **M1 (Week 2): Project Charter Signed**
  Scope baseline approved by academic mentors
- **M2 (Week 5): Database Schema Live**
  Books and reviews tables populated with seed data
- **M3 (Week 8): NLP Algorithm Alpha**
  VADER compound & RAKE keyphrase engine operational
- **M4 (Week 11): REST API Interoperability**
  FastAPI Swagger endpoints fully integrated with UI
- **M5 (Week 13): Advanced Analytics Lab**
  K-Means clusters, Apriori rules & social graphs live
- **M6 (Week 16): Final Defense & Handoff**
  Full code repository, documentation & live demonstration

IT PROJECT MANAGEMENT

CPM / PERT Critical Path & Activity Crashing Analysis

Three-time probabilistic estimation (T_e) identifying project bottlenecks and optimal crashing slopes.

PROBABILISTIC NETWORK

PERT Expected Time Formula

$$T_e = (a + 4m + b) / 6$$

a = Optimistic Time • m = Most Likely Time • b = Pessimistic Time

Critical Path Identification:

A → C → F → G → H (16.0 Weeks)

- Zero float activities; any delay in these directly delays project completion.
- Total Project Variance: σ^2 = 1.34 weeks (Std Dev: 1.16 weeks).
- 95% Confidence Interval for completion: 14.8 to 17.2 weeks.

SCHEDULE OPTIMIZATION

Crashing Cost Slope Analysis

Crash Cost Slope = (Crash Cost - Normal Cost) / (Normal Time - Crash Time)

- **Task C: NLP Pipeline Dev.**
  Normal: 4 wks → Crash: 3 wks • Slope: ₹2,000/wk (Priority 1 Crash candidate)
- **Task F: Dashboard Visuals**
  Normal: 3 wks → Crash: 2 wks • Slope: ₹2,500/wk (Priority 2 Crash candidate)
- **Task G: Integration QA**
  Normal: 2 wks → Crash: 1 wk • Slope: ₹3,000/wk (Priority 3 Crash candidate)
- **Task H: Final UAT Sign-off**
  Normal: 2 wks → Crash: 1.5 wks • Slope: ₹4,000/wk (High cost; avoid crashing)

IT PROJECT MANAGEMENT

Budget Breakdown (₹55,000 Cap) & Risk Register

Bottom-up cost accounting and quantitative risk management with proactive mitigation protocols.

FINANCIAL GOVERNANCE

₹55,000 Bottom-Up Budget

Strict capital expenditure cap verified against deliverable heads:

Software Engineering & Architecture	**₹25,000 (45.5%)**
Cloud Infrastructure & DB Storage	**₹10,000 (18.2%)**
Quality Assurance & Benchmark QA	**₹8,000 (14.5%)**
Miscellaneous & Contingency Buffer	**₹7,000 (12.7%)**
Technical Documentation & Reports	**₹5,000 (9.1%)**

RISK MITIGATION

Risk Governance Register

- **Review Noise & Spam** — P:Med | I:High
  Automated regex cleaning & duplicate token pruning
- **VADER Sarcasm Misses** — P:Med | I:Med
  Dual Naive Bayes fallback & compound threshold tuning
- **Database Data Drift** — P:Low | I:High
  Daily automated SQLite snapshots & foreign key constraints
- **Schedule Overrun** — P:Med | I:High
  CPM/PERT critical path tracking & crash candidate C
- **API Contract Mismatch** — P:Med | I:Med
  Pydantic V2 schema validation & Swagger documentation

VERIFICATION & RESULTS

System Verification & Academic Syllabus Compliance

100% formal compliance across all required BVDU CSBS syllabus units and live operational metrics.

< 10 ms
API Latency SLA
FastAPI Asynchronous Engine

89.2%
Supervised Accuracy
TF-IDF + Naive Bayes Benchmark

10 / 10
Aspect Coverage
Complete Domain Literary Reach

₹0 Cost Drift
Budget Variance
Exact ₹55,000 Cap Maintained

BVDU DET CSBS Academic Syllabus Compliance Matrix

Subject	Syllabus Unit / Concept	Practical System Implementation	Verification Status
ASTMA	Unit I: Text Preprocessing	Noise stripping, tokenisation, stopwords, lemmatisation pipeline	**Verified (100%)**
ASTMA	Unit II: Sentiment Analysis	VADER rule heuristics + supervised TF-IDF Naive Bayes benchmark	**Verified (100%)**
ASTMA	Unit III: Keyphrase Extraction	RAKE word degree to frequency co-occurrence matrix	**Verified (100%)**
ASTMA	Unit IV: Unsupervised Topics	K-Means TF-IDF space clustering (K=4) & Apriori association rules	**Verified (100%)**
Enterprise Systems	Unit I-II: 3-Tier MVC Architecture	FastAPI SOA REST gateway + Vanilla JS UI + SQLite persistence	**Verified (100%)**
Enterprise Systems	Unit III-IV: RBAC & TCO	Admin vs Customer role segregation & 5-Year COTS vs Bespoke model	**Verified (100%)**
IT Project Mgmt	Units I-VI: Project Controls	WBS, 16-Wk Plan, ₹55k Cost, Risk Register, CPM/PERT Crashing	**Verified (100%)**

CONCLUSION & ROADMAP

Project Achievements & Next-Generation Research Directions

Summary of engineering milestones achieved and future technological enhancements for 2026-2027.

DELIVERED VALUE

Engineering Deliverables

- **Full-Stack Enterprise Web Application**
  Production-grade 3-Tier MVC with FastAPI REST backend and responsive Chart.js dashboard.
- **Multi-Method NLP Opinion Engine**
  Real-time VADER valence scoring, RAKE keyphrase mining, and 10 literary aspect detectors.
- **Unsupervised Knowledge Discovery**
  K-Means semantic topic discovery (K=4), Apriori association rules, and reader bipartite graphs.
- **Rigorous ITPM Governance Framework**
  On-time 16-week delivery, probabilistic PERT critical path, and exact ₹55,000 budget adherence.

FUTURE HORIZON

Next-Gen Research Roadmap

- **Fine-Tuned Transformer Models**
  Deploy domain-specific Book-BERT or RoBERTa microservices for literary subtext and irony.
- **Sentence-Level ABSA**
  Syntactic dependency parsing to isolate multi-aspect polarities within complex compound sentences.
- **Vector Embeddings & Semantic Search**
  Vector database integration (FAISS / ChromaDB) for semantic recommendation search.
- **PostgreSQL Enterprise Clustering**
  Scale from SQLite to distributed multi-master database clusters with read replicas.